Splitwise (Expense Sharing System)

Patterns: Strategy Pattern, Factory Pattern, Greedy Graph Reduction | Domain: Finance / Expense Management

Core Requirements & Features	Core Domain Classes & Interfaces
• **User & Group Mgmt:** Create users, groups, and track net balance pairs across members. • **Expense Types:** Equal Split, Exact Amount Split, Percent Split. • **Debt Simplification:** Minimize total transaction count using greedy graph reduction.	• **SplitwiseService:** Facade managing users, expenses, & balance sheets. • **Expense & Split:** Represents expense payload & individual user share. • **SplitStrategy:** Interface implemented by Equal, Exact, & Percent strategies. • **DebtSimplifier:** Greedy min-heap / max-heap net balance resolver.

Critical Execution Workflow & Method Trace

1. Client invokes addExpense(paidBy, totalAmount, splits, SplitType).
2. SplitwiseService delegates validation & share computation to appropriate SplitStrategy.
3. Updates global user balance map: User.balances[debtor][creditor] += amount.
4. When simplifyDebts(groupId) is called, compute net balances for all group members: Net = Total Credit - Total Debit.
5. Separate users into Net Debtors (negative balance) and Net Creditors (positive balance).
6. Populate Max-Heap for Creditors and Min-Heap (or Max-Heap of absolute values) for Debtors.
7. Repeatedly pop top Debtor D and top Creditor C. Settle amount = min(|D|, C). Create simplified transaction (D -> C: amount). Update balances and re-insert if non-zero.

Concurrency & Thread Safety Mechanics	Design Trade-offs & Interview FAQs
• **State Locks:** ConcurrentHashMap> for balances. • **Race Prevention:** ReentrantLock per Group / Expense creation to prevent concurrent balance corruption. • **Atomic Updates:** DoubleAccumulator or atomic sync on user balance pair updates.	• **NP-Hard Simplification:** Optimal debt simplification is NP-hard (Subset Sum); Greedy O(N log N) yields near-optimal results. • **Precision:** Use BigDecimal instead of Double for currency arithmetic to avoid floating-point errors. • **Extensibility:** Adding new split type requires implementing SplitStrategy interface only.